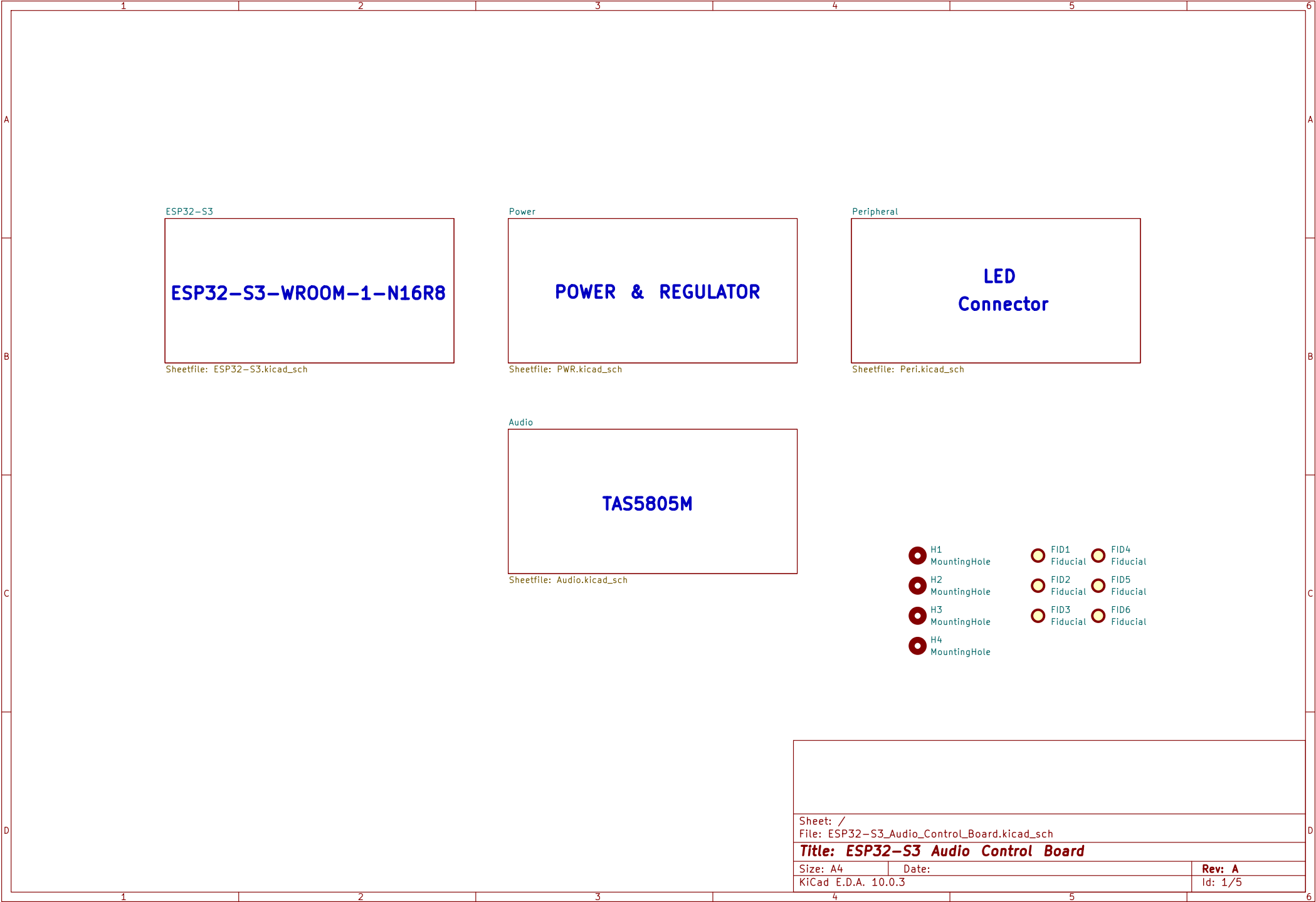
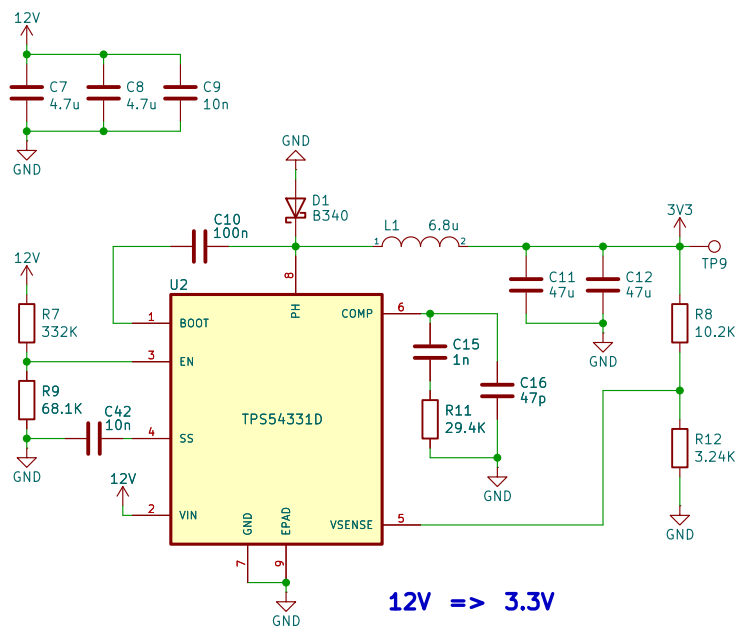
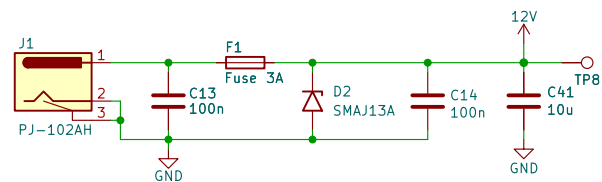




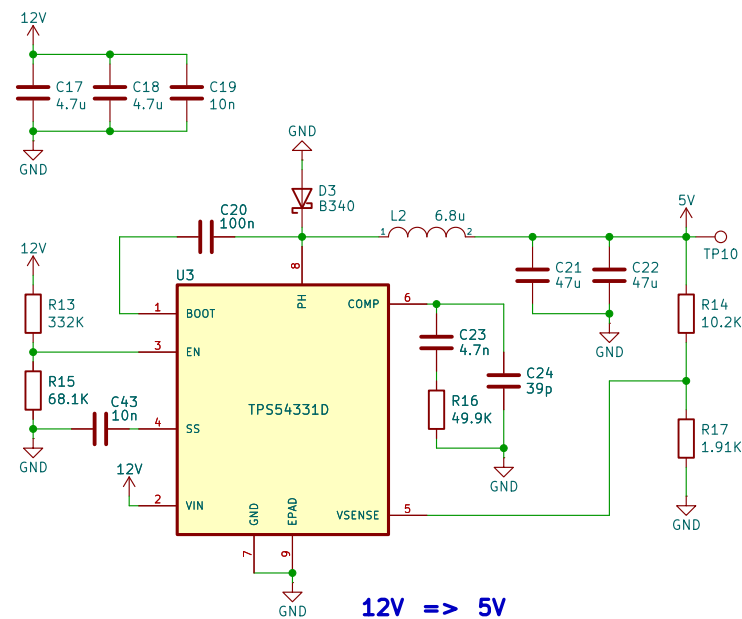


7.3.8 Current-Mode Compensation Design

To simplify design efforts using the TPS54331 device, the typical designs for common applications are listed in [Table 7-1](#). For designs using ceramic output capacitors, proper derating of ceramic output capacitance is recommended when performing the stability analysis because the actual ceramic capacitance drops considerably from the nominal value when the applied voltage increases. See [Section 8.2.2](#) for the detailed guidelines or use the WEBENCH software tool (www.ti.com/WEBENCH).

Table 7-1. Typical Designs (Refer to the Simplified Schematic)

V _{IN} (V)	V _{OUT} (V)	f _{sw} (kHz)	L _o (μH)	C _o	R _{O1} (kΩ)	R _{O2} (kΩ)	C ₂ (pF)	C ₁ (pF)	R ₃ (kΩ)
12	5	570	6.8	Ceramic 33 μF, ×2	10	1.91	39	4700	49.9
12	3.3	570	6.8	Ceramic 47 μF, ×2	10	3.24	47	1000	29.4
12	1.8	570	4.7	Ceramic 100 μF	10	8.06	68	5600	29.4
12	0.9	570	3.3	Ceramic 100 μF, ×2	10	80.6	56	5600	29.4
12	5	570	6.8	Aluminum 330 μF, 160 mΩ	10	1.91	68	120	29.4
12	3.3	570	6.8	Aluminum 470 μF, 160 mΩ	10	3.24	82	220	10
12	1.8	570	4.7	SP 100 μF, 15 mΩ	10	8.06	68	5600	29.4
12	0.9	570	3.3	SP 330 μF, 12 mΩ	10	80.6	100	1200	49.9



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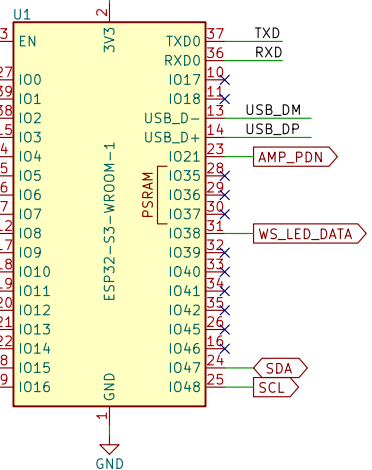
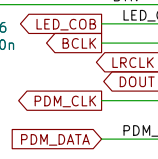
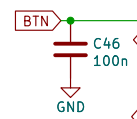
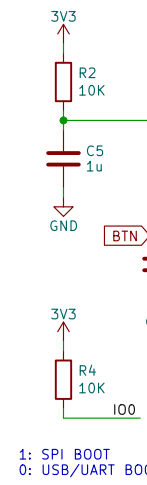
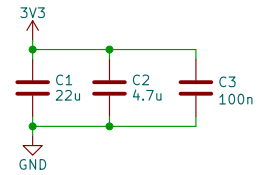
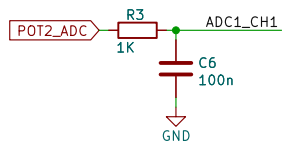
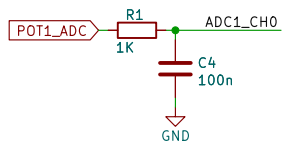
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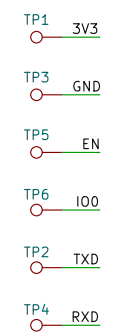
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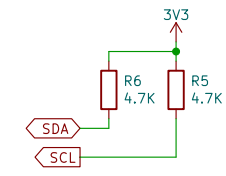
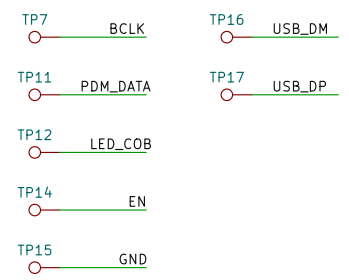
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UART



USB



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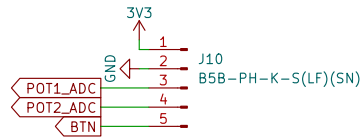
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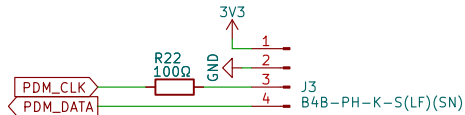
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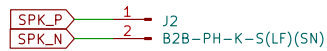
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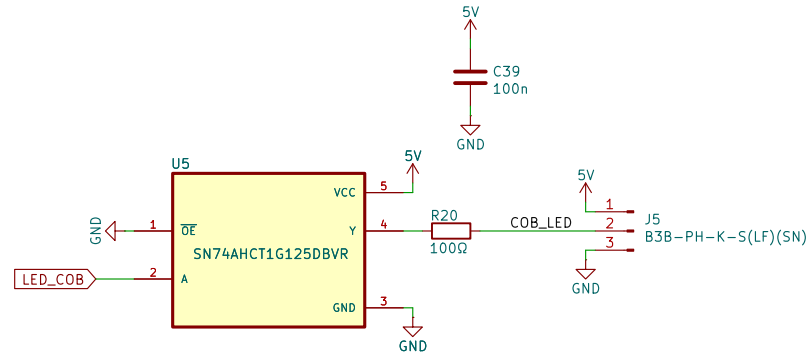
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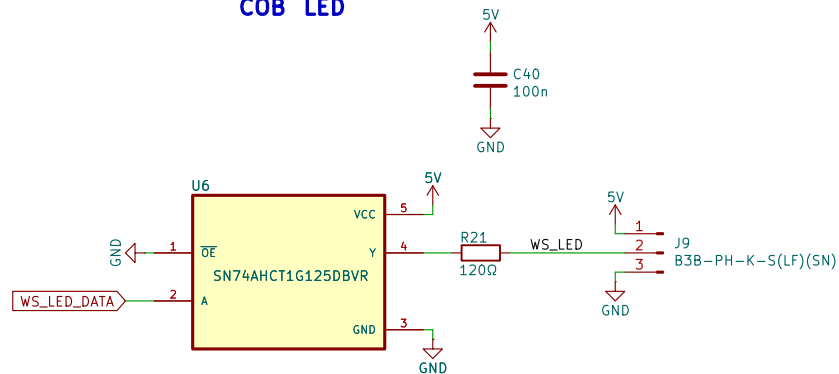
PDM Microphone



Speaker



COB LED



WS LED

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